
Verbalisation and the One-New-Idea Constraint

Naomi Peck, University of Freiburg, University of Bremen

Abstract:
The One-New-Idea-Constraint (ONICON) is proposed to be an absolute universal on human cognitive capabilities. While the ONICON predicts that we maximally realise one new idea at a time, it does not account for the maximal amount of information that can be realised at any point in discourse. To help us better understand factors influencing information flow in discourse, we may instead wish to turn our focus to verbalisation, the cognitive process behind this act. Examining verbalisation more closely can in turn allow us to better assess the extent to which the ONICON can hold. Using corpus data from three Trans-Himalayan languages, I explore how we choose to verbalise particularly information-rich intonation units and what factors may influence this process. Through investigating how information-rich intonation units are verbalised, I highlight ways in which the ONICON may be refined or challenged to better capture everyday language use.

Key words: One New Idea; information flow; verbalisation; Trans-Himalayan

1. Introduction

The search for linguistic universals has unearthed more ways in which languages are structurally different than they are similar (Evans & Levinson 2009).[1] What we instead recognise as universals in language stem from properties shared by humans and human communication worldwide. These universals can be interactionally-based, like the need to indicate what is being discussed (topics, subjects, cf. Chafe 1976; Givón 1983), or stem from constraints on the human body itself, such as the

[1] This article builds on thoughts from Chapter 7 in my dissertation (Peck 2025a), and as such, owes a lot to Uta Reinöhl and Walter Bisang. My thanks also go to the many who have discussed these ideas with me over the years, including Wifek Bouaziz, Laura Schleicher, I-Ying Lin, Pavel Ozerov, Stefan Schnell, Laura Becker, Guido M. Linders, and an anonymous reviewer.

limit on how much information can be produced at a single point in time, the so-called One-New-Idea Constraint (Chafe 1994).

The One-New-Idea Constraint (ONICON) is arguably the most well-known account of information flow. However, the limited empirical work available testing the constraint shows that a basic understanding of the constraint cannot account for the richness of how we encode information (Damron 2004; Linders & Schnell 2026; Ozerov 2026; Peck 2025a; Reinöhl & Peck 2026). As such, the constraint must be refined to address ways in which multiple new pieces of information can be realised in single intonation units, i.e. to account for INFORMATION-RICH INTONATION UNITS. While Chafe has already provided explanations for some information-rich intonation units (Chafe 1994: 110–118), he does not go into detail about the cognitive underpinnings for these accounts, nor does he discuss where the upper limit of information encoded in these intonation units may lie.

This article will explore potential factors underlying the distribution of these information-rich intonation units to better assess the extent to which the ONICON can capture the properties of information flow in naturalistic language. I go into the details of the ONICON in Section 2, before exploring the key cognitive process affected by this constraint, verbalisation, in Section 3. I introduce the data set I use to explore this question in Section 4. In Section 5, I look at four distinct factors which have been argued to affect verbalisation and how they are required to account for information-rich intonation units in a Chafian account. This is followed by a discussion of the limits of the ONICON in accounting for information-rich intonation units and a brief conclusion in Section 6.

2. The One-New-Idea Constraint

The ONICON is a universal put forward by Wallace Chafe, who claimed that we are limited as to how much new information we can realise at one time in an interaction (Chafe 1974, 1979, 1987, 1994). Chafe's key hypothesis is that our consciousnesses cannot handle more than "one new idea" at a time in an intonation unit, or a FOCUS OF CONSCIOUSNESS (Chafe 1994: 119). These ideas can be entities, events, or states, and are typically realised within discourse by content words. Crucially, ideas in Chafe's framework entail a cognitive cost to activate in the mind of an interlocutor (Chafe 1994: 80). This activation cost depends on how accessible we believe an idea is for interactants. The use of an inactive, or "new", idea within an interaction is hypothesised by Chafe to incur such a large activation cost that our consciousnesses are only able to focus on one of these at any given time, and therefore we are limited to realising a maximum of one of these in an intonation unit (Chafe 1994: 109). When we wish to introduce new information into an interaction, our consciousness must shift its focus accordingly.[2] This constant shifting of focus of consciousness from intonation unit to intonation unit is complemented by a more extended process of scanning at a higher level. This process makes use of semi-active information to contextualise the ideas encoded in intonation units with respect to each other and to our larger mental models in the formation of CENTRES OF INTEREST (Chafe 1994: 140–141). These centres of interest tie to larger organisations of prosodic information, such as sequences of intonation units, as well as to syntax-prosodic mismatches for the purpose of interaction (Chafe 1994: 142–143). As such, the ONICON is formulated as a constraint on the maximum amount of contentful information we can express in a single intonation unit, a unit which is inherently embedded within a larger discourse context.

The recognition of two relevant layers of cognition by Chafe highlights a gap in how we tackle information. Chafe argues that the realisation of a new idea in discourse represents the upper limits of what we can activate in our consciousnesses at any one point in time, and by extension, the upper limits that we can expect of our audiences (Chafe 1994: 109). However, Chafe remains silent on the upper limits of how much total information — both new and non-new — can be *encoded* in one of these foci of consciousnesses, the single intonation unit. It is possible for one new idea to be accompanied by other active or semi-active information, but how much additional information can

[2] Which exact cognitive mechanism is meant by "focus" is never explicitly spelled out by Chafe (cf. e.g. Chafe 1994: 79, 180, 2018: 17–18, 116, 127–128). I understand it as loosely corresponding to attention (see also Ozerov 2026).

be realised in an intonation unit apart from the new idea? In other words, the activation limit on new information should equal the maximum amount of cognitive load that we can expect an interlocutor to undertake at a given point in time but *not* the amount of information that can be realised in an intonation unit. We can thus say that the ONICON represents a constraint on cognitive processes in interactions, as opposed to a constraint on the expression of information.

If we take this representation of the constraint to be accurate, then we are left with the question of what drives the activation of information. The obvious answer is that information is activated due to interactional processes. These processes are driven by our wants to communicate information to one another, information which is typically derived from our personal experiences (e.g. Dunbar 2004; Szala *et al.* 2024). What, then, is the process that turns experiences into the information communicated in interactions? This is the domain of verbalisation. Verbalisation allows us to select a chunk of our experience and package it in a way that allows us to successfully communicate this experience to others. It is the act of turning thoughts into language by "taking a unique whole (the experience), and turning it into a sequence made up of reusable parts (the utterance)" (Croft 2007: 349). The act of "thinking for speaking" (Slobin 1996) requires us to construe experiences in certain ways which match common patterns in a given speech community. In other words, verbalisation is a cognitive process which mediates between units of experience and observable linguistic behaviour.

If the ONICON is a true universal of human cognition, we must look at where the limits of verbalisation lie. We are often confronted with intonation units in which multiple units of arguably "new" information are realised (cf. Chafe 1994: 110–118; Damron 2004; Peck 2025a: 370–392). It is exactly in these information-rich intonation units where we should be able to find gaps between the total amount of information realised and the amount of information which we can arguably activate at one time, and thus be able to more clearly observe the choices made to manage cognitive load in interactions. That is, we must look at instances where the new information expressed in single intonation units may hit the upper limit of what could be considered “too much”. These information-rich intonation units are key for gaining a deeper understanding of the ONICON and assessing whether it accurately captures processes of information flow.

3. Verbalisation as a cognitive process

For such an important process for language and cognition, verbalisation is relatively understudied. Theories of language production invoke verbalisation in early stages within their models, but often do not discuss the process of how experience is deconstructed into a form which can be expressed in language (Croft 2007: 346–348; Ünal, Ji & Papafragou 2021: 226). That is, they begin from the point that a cognitive unit is determined and do not go into detail about how that unit is selected from the larger stream of experience (e.g. Levelt 1989; see Papafragou & Grigoroglou 2019 for a recent review). Approaches to investigating cognition which leverage verbalisation often take a similar position, in that the internal process of verbalisation is not considered in much detail. In these works, the process is simply taken to be an act which results in a level of cognitive load on the speaker, depending upon the amount of active attention needed to formulate an utterance, with little attention paid to the factors which determine linguistic choices (cf. e.g. Ericsson & Simon 1980, 1998; Tenbrink 2015; Vogels *et al.* 2020). A fuller understanding of information flow requires us to take a closer look at the inner workings of verbalisation, from thought to final expression, to properly understand the effects that cognitive load may have on language choice.

For the purposes of this article exploring the ONICON, I borrow the basic framework of Croft (2007)’s model of verbalisation, which takes Chafe (1977)’s model of verbalisation as a starting point, but equips it with further steps which turn the selection of content words into grammatically-acceptable utterances. Croft (2007) takes verbalisation as encoding the proposition within a speech act (2007: 347; cf. Clark 1996), with the speaker presenting the proposition and the addressee as identifying the proposition. He splits the process of verbalisation into six distinct stages: subchunking, propositionalising, categorisation, particularising, structuring and cohering. These processes occur to some extent simultaneously and represent the act of utterance production from conceptualisation through to just prior to articulation (cf. Levelt 1989). This contrasts with Chafe's more recent model of verbalisation (Chafe 2018), who explicitly builds phonological mapping into his model and collapses many of the stages that Croft distinguishes in his workflow. An overall schematisation of Croft's model can be seen in Figure 1.

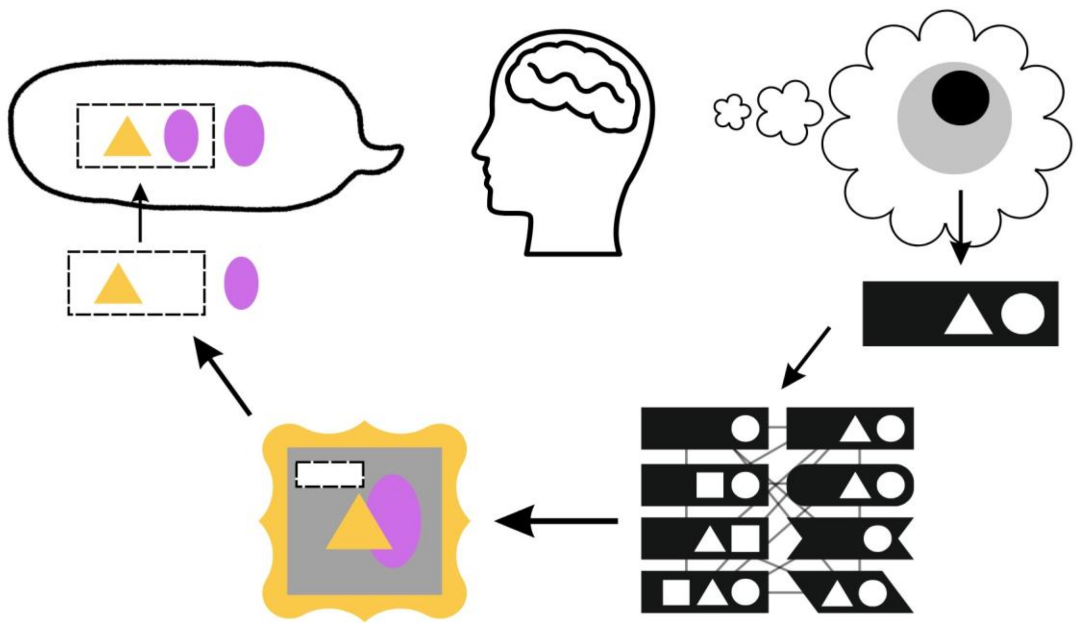

Figure 1. Visualisation of Croft's model of verbalisation

We begin the process of verbalisation with choosing what we want to talk about. Subchunking in Croft's model represents the act of selecting the experience to be verbalised and choosing a part of this whole to be further processed (Croft 2007: 349). This process is portrayed as the breaking down a whole to its constituent subparts by Croft (2007: 349, 354). In contrast, the same process is presented as "zooming in" on the experiential whole by Chafe, who portrays the selected subpart as constituting the focus of consciousness and its surrounding environment as constituting the periphery of consciousness, or the centre of interest (1994: 140–143). This stage best corresponds to selection in

Chafe's updated model of verbalisation (2018: 72) and corresponds to what is known more commonly elsewhere as the act of event segmentation (cf. Zacks & Swallow 2007).

The subchunk selected to be verbalised then undergoes propositionalising. At this point, the slice of experience is transformed into a preverbal proposition where the speaker chooses what to individuate (entities) and what not (events) (cf. e.g. Chafe 1998). Already at this stage, we can see effects of language on how a subchunk is propositionalised (Slobin 1996; Trueswell & Papafragou 2010). This stage is incorporated under categorisation in Chafe's more recent understandings of verbalisation; however, the act of individuation is assumed to be part of categorising parts of the experience as entities or events, as opposed to a precursor for the recognition of a cognitive unit as an event or entity for categorisation. The choice of which entities to individuate within the experience leads to this stage also being termed perspectivising (e.g. Stathi 2023: 19).

The last stage in Croft's model to be adopted from Chafe (1977) is categorisation, which involves the beginning of linguistic encoding. In this stage, the proposition from the previous stage enters grammatical encoding, whereby the entities and events are categorised based on previous experiences of the speakers and suitable linguistic mappings are found for the relevant concepts (cf. Malt *et al.* 1999). While Croft portrays the result of this stage as a collection of contentful lexical items which can express the cognitive ideas present in the preverbal message (2007: 351), I prefer to interpret the stage in line with Chafe's most recent account of categorisation. That is, the result of categorisation is a collection of discourse ideas which can be linked to linguistic categories, but no distinct linguistic form for these ideas have been determined as of yet (cf. Chafe 2018: 74).

Although appropriate discourse ideas have been identified for the verbalisation of the given subchunk of experience, they still lack lexical realisations and distinct features which would allow listeners to reconstruct the intended preverbal message upon their verbalisation. That is, the result of categorisation is idea "types", rather than "instances (of types)" or "grounded (instances of types)", i.e. tokens of the ideas (cf. Langacker 1991: 51–55). This motivates the second half of Croft's verbalisation model. The fourth stage of verbalisation, particularising, involves selecting tokens (lexical items) for the more general type (discourse ideas) and situating these tokens within the discourse world. Selecting involves individuating the particular entity or event under discussion, quantifying it, and distinguishing it from other potential members of the type set (Croft 2007: 358). In comparison to what occurs at the propositionalising stage, individuation at this stage of verbalisation can be understood as the act of choosing an appropriate expression for the particular entity or event that distinguishes it from other instances belonging to the same discourse idea type. Situating involves taking these selected particulars and locating them relative to established figures and grounds within the discourse world. It also involves the speaker situating, or orienting, themselves relative to the information being verbalised (Croft 2007: 363). Croft's notion of situating corresponds to Chafe's notion of orientation as expressed in Chafe (1994), whereby speakers orient themselves solely to the information being encoded without taking into consideration the wider interactional context (Croft 2007: 363). However, I am more sympathetic to Chafe's later expressions for this stage, whereby the activation state of interlocutors factors into lexical choices made during orientation (e.g. Chafe 2018: 75). As lexical items arguably decay quicker in working memory than discourse ideas (Yule 1981), choices about how to verbalise a discourse idea at this stage of the process must necessarily involve assumptions about how active a given discourse or lexical idea is in an interlocutor's consciousness. More broadly, lexical choice at this stage of verbalisation has also been linked to other ways in which a speaker's choices can make understanding easier for an active interlocutor (e.g. Grigoroglou & Papafragou 2019). This interlocutor-centric perspective results in more effective particularising overall on the part of the encoder, rather than waiting until the final stage of verbalisation (cohering) to choose whether a pronoun should be used or not. The final output of this stage of verbalisation are lexical items which should be correctly interpreted by an audience as encoding discourse ideas.

The last two stages of verbalisation involve the finer choices involving grammatical encoding. The structuring stage of verbalisation takes the selected lexical items and plugs them into appropriate grammatical structures. This is followed by the stage of cohering, which edits these forms to make them pragmatically acceptable through relating them to other units in the ongoing discourse context. Both of these stages correspond to Chafe's literalisation, the only process he posits that turns semantic structure into syntactic structure (Chafe 2018: 76–78). The choice of structures at both of Croft's stages is highly dependent on the range of factors which affect how we construe experiences (see Section 5). As these two stages primarily affect functional elements, as opposed to contentful items, I will not discuss them further in this article.

The end result of the verbalisation process is a linguistic expression of a slice of experience which is transformed by cognitive processes (Chafe 1977, 1994, 2018; Croft 2007: 377). Verbalisation necessarily involves decisions on the part of the speaker as to how to construe the experience: what slice of the whole is presented as a subchunk, which entities are individuated when propositionalising, how the parts of the proposition are related to previous experiences, how these parts are particularised within the discourse world, how the particularised proposition is structured, and how this is converted to fit the ongoing interactive situation. Each stage of the verbalisation process involves active construal, or active decisions on the part of the speaker about how to portray the selected slice of experience.

As it stands, there are some uncertainties of how to implement and interpret this view of verbalisation and the ONICON. The process of verbalisation creates a (minimally) three-way distinction of relevant levels of information. We begin dealing with cognitive ideas at the level of subchunking, discourse ideas as the output of propositionalising, and lexical items as the output of particularisation. In a Chafian approach to information, thoughts and their verbalisation are treated as separate phenomena with many-to-many relationships. However, Chafe appears to often equate the lexical items used in discourse with the cognitive or discourse units in the mind of an interactant, making it unclear exactly how to envisage what kind of information is being discussed or how this may be affected by verbalisation (see also critique in Delin 1996; Reinöhl & Peck 2026). It is also unclear to what extent we can separate out cognitive and discourse ideas, as we partially experience the world via language (e.g. Hopper 1995: 149; Wittgenstein 1953: sec. 580). Furthermore, as verbalisation is shaped by an interactant's belief about the state of information in the consciousness of their audience (Chafe 1994: 75), the realisation of an idea or centre of interest cannot necessarily reflect the status of this information in the interactant's consciousness, although we do make attempts to reflect it in our verbalisation (Grigoroglou & Papafragou 2019; however, cf. Vogels *et al.* 2020). If the ONICON truly pertains to cognitive load, then it should theoretically affect all stages of the process which involve the active management of meaningful cognitive resources, extending from the size and granularity of subchunk selected through to the type of contentful lexical items chosen.

There is also little discussion of how language comprehension may work in these approaches to verbalisation, except as the same process happening in inverse (e.g. Chafe 2018: 128–130). If true, this would mean that we would expect cognitive load to occur beginning at the stage of decoding the particularisation encoded by an interlocutor. This stage would be where an interlocutor would have to begin reconstructing the "transformations" to the original utterance, leading to the imperfect reconstruction of the original experience by the audience. There is, however, a clear preference within the comprehension or "reconstruction" process for the subchunked unit of experience, as information associated with event boundaries is generally more easily retrieved than information within the middle of an event (e.g. Swallow, Zacks & Abrams 2009). This suggests that parsing may to some extent reflect an inverse process of verbalisation, although language processing may arguably be "greedier" than the process of language production (cf. Christiansen & Chater 2016).

4. Case study

To understand where the limits of verbalisation are, we must look at instances in which the ONICON may feasibly not hold. What are the conditions which allow for the realisation of information-rich intonation units? This requires us to look beyond single intonation units to the wider discourse context to investigate how information is realised in naturalistic data.

Using a Chafian approach to investigate information requires distinguishing lexical and discourse units from cognitive units. While it is relatively easy to compare and contrast lexical items, no annotation scheme of discourse units reliably captures all the different types of cognitive ideas that Chafe discusses (see discussion in Reinöhl & Peck 2026). To identify and classify discourse units, I used TONIC (**T**esting the **O**ne-**N**ew-**I**dea **C**onstraint, Reinöhl & Peck 2026) in combination with ActIND (**Act**ion **I**ndexing in **N**atural Language **D**iscourse, Peck 2025a), an annotation scheme which tracks identity of event and state ideas. As both of these schemes do not rely on syntactic criteria to identify discourse units, they allow us to break free of the circularity common to studies of eventhood (cf. Givón 1991; Pawley 2011).

This multi-pronged approach lends itself nicely as a base for investigating the relationship between lexical, discourse, and cognitive units. By tracking the flow of ideas within discourse, we can hypothesise as to their activation status at the time of utterance. When an intonation unit only contains a single new piece of information, we can hypothesise that this unit corresponds to "one new idea" and thus potentially a single cognitive unit. If an intonation unit contains multiple pieces of information which were annotated with "new", then the intonation unit represents an instance in which we can explore what mechanisms allow for such an information-rich unit to be realised.

For this investigation, I reuse the discourse subcorpus explored in Peck (2025a). This corpus consists of three texts of comparable length and genre with thick annotations in three different languages of the East Himalaya (Kera'a, glottocode: idum1241; Duhumbi, glottocode: chug1252; and Galo, glottocode: galo1242). These texts are Dog Story (Kera'a, Peck 2025b), Macaque (Duhumbi, Bodt 2018a, 2018b), and Lost Writing (Galo, Post 2006). A full description of the stories and their contents can be found in Peck (2025a: 429–443) and ELAN annotations for these files can be found at https://osf.io/tmxzp/. The annotations for these files specifically focus on the information status of events and states (hereafter, just referred to as events). These texts had a total of 261 unique discourse event ideas, which were realised 467 times.

To find the limits of what our consciousnesses can handle, we first must establish the limits of the "basic" form of the ONICON. That is, we must see to what extent the first realisations of discourse ideas are the only pieces of new information in an intonation unit. These results can be seen in Figure 2. Roughly half of all new event ideas were the only piece of discourse-new information in an intonation unit. However, a substantial number of new events in the three texts co-occurred with other pieces of new discourse information in the same intonation unit, primarily other events. A manual inspection of the information in the texts also found no additional instances where more than one new referent idea occurred in the same intonation unit above and beyond what is captured in the investigation of event ideas. The results depicted in Figure 2 means that a simplistic understanding of the ONICON cannot cope with a decent amount of naturalistic discourse (cf. Damron 2004), a conclusion already reached in Chafe (1994).

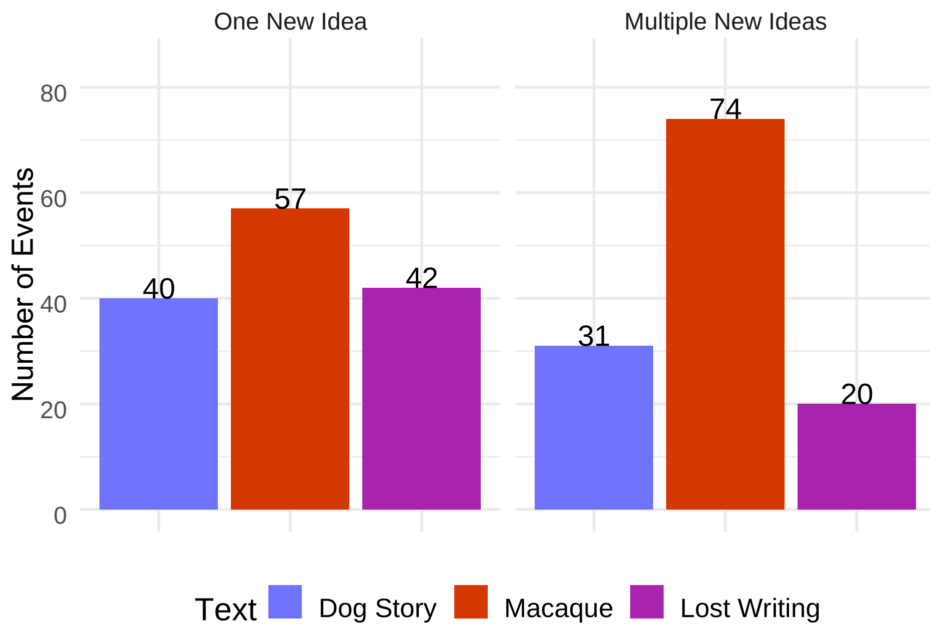

Figure 2. Number of new ideas in an intonation unit

The distribution of results in Figure 2 suggests a more nuanced version of the ONICON is needed to account for information-rich intonation units. Chafe argues that conventionalised expressions, low-content discourse ideas and/or lexical items, and previous activation of cognitive ideas allow us to encode more information at a single time (Chafe 1994: 108–119; cf. also Croft 1995; Pawley & Syder 1983, 2000; Reinöhl *et al.* 2023). These categories correspond to multiple pieces of information which constitute a single idea, multiple pieces of information which map to multiple ideas, of which one is low in activation cost, and multiple pieces of information that correspond to separate ideas that have been activated at different times. Reinöhl *et al.* (2023) argue for positing additional scenarios to account for complex lexical expressions, such as semantic overlap (see also discussion in Reinöhl & Ellison 2026; Reinöhl & Peck 2026). We may additionally argue that speakers "pre-activate" cognitive ideas by using certain grammatical structures such as contrastive focus (Peck 2025a: 383–384; cf. also Chafe 1974: 117–119). However, these explanations focus on the distribution of information, rather than the process of encoding the information itself. In order to identify potential factors affecting verbalisation, we must look more in detail at these information-rich intonation units to see what may be at play.

5. Factors affecting verbalisation

Cognitive units, just like the process of verbalisation itself, cannot be directly observed. We can only guess at what might be happening in our consciousnesses by inspecting the various factors affecting how we choose to verbalise a given cognitive unit. Croft argues that the main cognitive process affecting verbalisation is construal (2007: 377), and that this process is jointly determined by cultural conventions, interactional goals, and the "nature of reality" (Croft 2007: 350). This would suggest that different ways of construing experience lead to the kinds of information-rich intonation units that a "basic" form of the ONICON cannot account for. Applying Croft's insights as well as Chafe's original accounts to my dataset, I tentatively suggest that a slightly different grouping of factors determining construal may be warranted.

In the following sections, I look at four different ways in which we can explain information-rich intonation units via construal affecting verbalisation: conventionalisation (Section 5.1), world knowledge (Section 5.2), intersubjectivity (Section 5.3), and interactional goals (Section 5.4). These four factors help us in concert to decide how to verbalise the ideas we wish to communicate. In these sections, I explore examples which are particularly illustrative for each of the factors; however, whether these factors represent an exhaustive account of relevant factors affecting verbalisation and how these factors interact with each other await further scrutiny.

5.1 Conventionalisation

A number of instances in which the ONICON may appear to not hold correspond to instances in which a speaker chooses to use multiple lexical items to construe a single discourse unit (cf. Chafe 1994: 113–116). The construal represents a choice by a speaker to opt for a linguistically conventionalised expression, or “prefab”, in the verbalisation of a given experience. That is, during the particularising stage of verbalisation, the speaker opts to encode a discourse idea using multiple lexical items which recognisably belong together as a single functional unit (see also Reinöhl & Ellison 2026). This falls under Croft’s notion of “cultural conventions”, in that prefabs represent a conventional understanding of multiple lexical items as representing a single discourse idea in a given speech community.

Relatively few instances of information-rich intonation units within the dataset could solely be explained via prefabs proper. The most common instance was when a discourse-new referent co-occurred with a discourse-new event, which occurred primarily with light verb constructions, like in (1–2).[3] In these examples, *zok* ‘lie’ and *le* ‘do’ are both discourse-new information which may correspond to separate discourse ideas. However, it is the combination of the two elements which results in the discourse events associated with `043` and `077`, the bat lying to the macaque (on multiple different occasions). Furthermore, the phrase *zok le* is arguably frequent enough in Duhumbi that the co-occurrence of the two lexical items can be interpreted as encoding a single idea rather than multiple ideas.

(1) Duhumbi: Macaque 61[4]

gomtɕʰen pʰawaŋ ***zok leda*** *wojɲi*

gomchen_phawang	zok	le-da	woy-nyi
bat	lie	do-NF	3SG-DAT
1	new	new	2
new	new	new	pro

[3] Examples follow the following conventions: The first line is a broad IPA transcription. Punctuation on this line represents notable pitch movements, with a full stop representing a fall, a comma representing a rise, and a double slash representing a terminal intonation contour. Bars (|) represent intonation unit breaks, with multiple bars representing stronger breaks. The second and third line encode grammatical units and their glosses, respectively. The fourth and fifth line represent d-level and l-level annotations as per TONIC (Reinöhl & Peck 2026). The sixth line contains ActIND annotations, which take the form of three-digit indices.

[4] Glossing follows the standard Leipzig glossing conventions. Non-standard glosses are as follows: ADD additive, ANA anaphoric pronoun, CC clause chainer, CMPL completive aspect, COP.LE *le* copula (Duhumbi), CV converb, EQ equative copula, EXIST existential copula, HORT.POL polite hortative marker, NF non-final, OK tag particle, SBRD subordinator, SEQ sequential marker, SUBJ subjunctive, WHEN ‘when’ subordinator.

043
'(that) bat telling him (the macaque) a lie…'

(2) Duhumbi: Macaque 94

zok leda *k^{h}um bojuba*

zok	le-da	khum	boyu-ba
lie	do-NF	eye	NEG.EQ-NMLZ
new	new	5	new
new	new	2-same	new
	077		078

'…(the bat) telling that lie, (fed to the macaque) not the eye…'

A superficially similar instance can be seen in (3), where the event 006 (the bat staying eating) is associated with two verbs. The verb *ju* 'stay' is used here by the speaker to encode the continuity of the eating action rather than a separate act of staying. The aspectual category is often verbalised with verbal elements used elsewhere in serial verb constructions. As such, the use of *ju* in (3) can be understood as conventionalised, even if it does not constitute a set prefabricated expression.

(3) Duhumbi: Macaque 13

tɕ^{h}ik^{h}us ***kada dʑuloŋbeʔ*** *, //*

chikhus	ka-da	ju-long=beq
honey	chew-NF	stay-PRF=EXIST
1	new	
1-same	new	new
	006	006

'(The bat) had been eating honey.'

However, it is not necessarily the case that *ju* must be realised in the same intonation unit as the action it modifies. (4) shows an instance of 033 (the bat staying eating) in which the verbs associated with the act of eating and the act of staying were realised in different intonation units. This shows that conventionalisation alone cannot explain all cases of information-rich intonation units. Other factors must also play a part in the particularising process above and beyond conventionalisation.

(4) Duhumbi: Macaque 51

gomtɕʰen pʰawaŋ dapda | ogi tɕʰikʰus || ***kada*** || ***dʑubale*** *. //*

gomchen_phawang	dapda	ogi	chikhus	ka-da	ju-ba=le
bat	again	that	honey	chew-NF	be-NMLZ=COP.LE
16		55		new	1
new	1-same	pro	new	new	new
				033	033

‘Again the bat stayed eating that honey there.’

5.2 World knowledge

The next factor which affects construal is shared world knowledge. A number of examples involve the activation of multiple units of information which arguably belong to a single cognitive schema. Cognitive schemas form part of the long-term knowledge associated with mental models that interlocutors use when processing input (Radvansky & Zacks 2014: 27–28; cf. Johnson-Laird 1983). These schemas are based on knowledge derived from language as well as personal experience and can represent schemas based on (presumably universal) human experiences as well as more culture-specific happenings. Accounting for shared world knowledge affects all stages of verbalisation which involve the manipulation of cognitive units, from propositionalising to particularising. However, relevant schemas are probably activated as early as the subchunking stage, where a speaker decides on their centre of interest (cf. Chafe 1987: 29–31).

World knowledge is arguably similar to conventionalisation, in that both represent knowledge stemming in part from “cultural conventions” (Croft 2007: 350; cf. also Chafe 1994: 111–118). I distinguish these factors for two reasons. The first is that there is a difference as to when both factors affect verbalisation. While world knowledge may affect multiple stages of the verbalisation process, conventionalisation is only relevant during particularisation. Secondly, while conventionalised expressions are directly sourced from knowledge of linguistic use, not all cognitive schemas implicated in world knowledge must come from linguistic experience and/or conventions in a given speech community. For example, some cognitive schemas arise via processes such as embodiment.

A clear instance of an embodied cognitive schema at work can be seen in (5). In this example, all elements in the second intonation unit are discourse new. However, the state of blindness encoded by 089 can be understood as a natural result of the events encoded in the prior intonation unit. In this case, we can argue that the verbalisation of two new ideas in the second intonation unit is permitted due to the activation of a schema built on embodied knowledge, namely, that eyes are used to see. Even if someone has not experienced the loss of their eyes directly, they can imagine that losing their eyes would result in the loss of sight, or turning blind. The activation of the cognitive schema surrounding sight allows for the verbalisation of an information-rich intonation unit.

(5) Duhumbi: Macaque 101

ose leda waʔ kʰumna ɲiska brukta madabale . | ***kʰum kana ɕetoŋbalebej*** . | *bi woj hoʔ*

|| *baŋbale kʰum* .

ose	le-da	wa-aq	khum=na	nyis-ka	bruk-ta
like_that	do-NF	3SG-GEN	eye=FOC	two-both	dig-NF
	1	[6]	5		17
	new	pro	3-same	5-same	4-same
	087				074

ma-da-ba=le	khum	kana	shet-oŋ-ba=le=beq=aj
finish-keep-NMLZ=COP.LE	eye	disabled	exit-go-NMLZ=COP.LE=EXIST=OK
new	new		new
new+new	1-same	new	new+new
088			089

bi	woy	hoq	bang-ba=le	khum
ANA	3SG	now	NEG.EXIST-NMLZ=COP.LE	eye
1	2		new	3
pro	pro		new	2-same
			090	

‘Doing like this, digging out both of his eyes, (they) finished. So he (the macaque) turned blind. He had no eyes.’

We can also find examples of schemas which may be culture-specific. Eating events form a part of a potential schema in (6) from Galo. In this instance, the action of roasting the hide with writing on it (013) is sequentially followed by the act of eating the hide with rice (014). This is motivated by the idea that *acin* ‘rice’ should be eaten with an accompaniment (011). Rice is typically eaten for every meal by the Galo, accompanied minimally by green leafy vegetables (Baruah & Bhattacharyya 2019: 198). When meat is to be eaten on the same day as part of the meal, it is typically prepared by either roasting it over a fire or steaming it by wrapping it in a leaf and burying it in the ashes of a fireplace (Baruah & Bhattacharyya 2019: 199). As such, all of the new idea annotations in (6) likely reflect parts of a commonly shared cultural schema surrounding the make-up and preparation of meals. This could have led to the speaker choosing to include multiple units of "new" information in the same intonation unit, as they would assume that the activation of the cognitive schema by their interlocutor would allow them to deal with more information than normal in a single intonation unit. That is, the world knowledge attached to this cultural schema would account for why multiple units of new information were realised in the same intonation unit — because they were not truly inactive at the point of their verbalisation.

(6) Galo: Lost Writing 12

apin əm |||| ***atɕin domᵊ*** *naːn kaːmaː leːmə* ||| ***baːl dokaː ku*** . //

apìn=əm	acín	dó-mə́-náa-nà	káa-máa-lèe=əm=əə
skin=ACC	rice	eat-COM-INSTR-SBJ,NMLZ	have-NEG-SEQ=SBRD=TOP
1	new	new	new
4-same	new	new	new
		011	012

báa-là	dó-káa-kú
bake-NF	eat-PRF-CMPL
new	new
new	1-same
013	014

'…having nothing to eat with their rice, (they) roasted and ate (their) hide (that had been given to them).'

5.3 Intersubjectivity

The third factor which I argue affects construal is intersubjectivity. Here, I take intersubjectivity to be assessments by speakers of what information their fellow interactants have access to (cf. Evans, Bergqvist & San Roque 2018: 111). This factor permeates all stages of the verbalisation process, but is most relevant at the stage of particularising. Croft (2007) does not discuss intersubjectivity explicitly; however, I believe it is worth addressing separately due to the nature of how Chafe models activation. Intersubjectivity is key to the ONICON: Chafe argues that we limit ourselves to activating only so much information at one time based upon our beliefs about the state of consciousness of our interlocutors (Chafe 1994: 109). As such, intersubjectivity is an important process which affects how we construe ideas, and therefore also how we verbalise them.

One way in which information-rich intonation units can be explained via intersubjectivity is by the "typicality" condition. That is, the speaker assumes that by explicitly verbalising one discourse idea, their interlocutor will activate this idea in addition to another discourse idea. This often requires shared cognitive schemas to be effective, but may also rely on the speaker using certain constructions to achieve their interactive goals (cf. Chafe 1994: 114). However, this process crucially relies on the notion of newsworthiness to work: information which is typical for a given idea can go unrealised, while atypical information should be realised, as this information is newsworthy for the interlocutor (Mithun 1987; Papafragou & Grigoroglou 2019: 1123). This results in typical information being construed as discourse-given and being particularised accordingly.

An instance in which we can see intersubjectivity actively at work is in information-rich intonation units involving quotation. The attribution of a quote to a referent or the reporting of a speech event commonly involves multiple new events occurring within the same intonation unit, such as in (7) from Kera'a. In (7), the verb *la* 'say' is used to encode a speech event alongside the quoted material. Multiple quotes can also occur in the same intonation unit, as in (13). The realisation of a quotation entails the presence of a "quoter", i.e. a narrator who says the quote, and an act of "quoting", i.e. a saying event. Even if the quotation itself is not accompanied by these two concepts explicitly, speakers assume that their audience activates this frame along with the quotation and this information enters the common ground of the conversation.[5] We can thus argue that speakers construe the act of

[5] For an example showing this in Kera'a, see Peck (2025a: 465).

quoting as being "given" when accompanying a quotation, based upon their assumptions about the consciousnesses of their interlocutors (see also Chafe 1994: 112). Quotation events of this nature have been previously argued to be "low-content" ideas elsewhere (Chafe 1994: 113; Peck 2025a: 379–380).

(7) Kera'a: Dog Story 35

ai || ***atʰu ana laho***

ayi	athu	a-na	la-ho
like_this	see	like_that-IRR	say-WHEN
	new		new
	new	new	new
	018		019

'(the dog) said take a look …'

Intersubjective construal appears to go beyond the simple modelling of the current "focus" to the modelling of more complex cognitive processes by speakers. Consider (8). Prior to this example, the speaker established the characters of the foolish macaque and trickster bat, and the location of the two characters as nearby a beehive. She goes on to realise two discourse-new pieces of information, *gepuaq chiling* 'king's drum' and event 002 (the bat guarding the king's drum), in the second intonation unit. Upon first encounter, the content of this intonation unit appears to be tangential and unrelated to the main story (especially as there is no clear reason for a king to be relevant for the tale). However, we must believe the speaker utters this intonation unit in good faith and thus it must be relevant for the tale at large (cf. Grice 1975).

(8) Duhumbi: Macaque 31

gomtɕʰen pʰawaŋ | *lok ga* ***gepuaʔ tɕʰiliŋ dʑakda*** || *inda inloŋbeʔ*

gomchen_phawang	lok	ga	gepu-aq	chiling	jak-da
bat	friend	1SG	king-GEN	drum	guard-NF
1	4	1	[new]	new	new
new	4-same	pro	new	new	new
					002

in-da in-long=beq

say-NF say-PRF=EXIST
new
3-same 0-same
020 020
'The bat had said, "friend, I guard the king's drum."'

For the ONICON to hold, the information-rich intonation unit must be explained via inference (cf. Lambrecht 1994: 101). The verbalisation of such a unit implies that the speaker believes some of this information must be activated in her interlocutor's consciousness. At least one of the two discourse ideas could feasibly be linked to activated ideas via the inference of three facts. First, an interlocutor should know or be able to infer that *gepuaq chiling* 'king's drum' is similar in size and shape to a beehive.[6] Secondly, the discourse-given action of the bat staying near the beehives should be similar in nature to the action of the bat guarding something valuable, such as a king's drum (002). Third, lying is something that the bat is motivated to do as a trickster character. Inferring these three facts would result in a correct interpretation of both *gepuaq chiling* and 002 as being lies said by the bat and thus false representations of previously established facts about the discourse world. The utterance of (8) represents an instance in which a speaker assumes that an interlocutor will be able to infer these facts based upon information which is available to them. This information could only arise a situational model in which referents with primary importance in discourse (cf. Chafe 1994: 88–89) are modelled with information about their interactional goals (Graesser, Singer & Trabasso 1994; Magliano & Radvansky 2001). Furthermore, the speaker would have to assume that their audience would actively model this information or predict that they would model this information via forward modelling (Pickering & Garrod 2013).

Intersubjectivity can also explain the co-occurrence of multiple new pieces of information in (9). This example involves the realisation of four discourse-new elements in one intonation unit: *khum nyiska* 'both eyes', 074 (both of the macaque's eyes being dug out), 075 (the macaque's eyes being

[6] A quick Google Images search of Tibetic drums and beehives certainly suggests that this may be the case.

thrown away), and 076 (the macaque not seeing). The three events follow (from) one another: 074 is followed by 075, and 076 is construed as the result of these two events.

(9) Duhumbi: Macaque 92

waʔ k^{h}um ɲiska brukta toshatda k^{h}um baajbalebej bi .

wa-aq	khum	nyis-ka	bruk-ta	tos-hat-da
3SG-GEN	eye	two-both	dig-NF	throw-lose-NF
4(1)	new		new	new
pro	2-same	new	2-same	new+2-same
			074	075

khum	ba-ay-ba=le=beq=aj	bi
eye	NEG-see-NMLZ=COP.LE=EXIST=OK	ANA
0	new	0
0-same	new	pro
	076	

'Both of his two eyes dug out and being thrown away, it's so that he (the macaque) wasn't seeing (a thing), ok.'

Of these discourse-new ideas, three of the four had arguably been previously activated. As the causal result of 074 and 075, 076 can be argued to be activated by an appeal to the same embodied cognitive schema around blindness which accounts for (5). If we argue that speakers actively take situation models maintained by their audience into account during the construal process, we may also consider 074 and *khum nyiska* 'both eyes' as previously activated if the wider interactional context is considered. At this point in the text, the bat has already dug out one of the macaque's eyes, and the immediately preceding intonation units discuss the bat digging out the second eye. If speakers assume that interlocutors are actively maintaining and updating a situation model of the discourse, they may assume that these or related ideas already are semi-active in the centre of interest. However, 075 cannot be accounted for in the same manner: it is not immediately predictable based on the prior discourse context, nor is any event involving the throwing away of eyes realised in the text prior to this point.

There are two possible explanations for `075` being realised alongside other activated information that are compatible with the ONICON. The first explanation is that `075` truly constitutes discourse-new information and that it is possible to realise both inactive and semi-active ideas in the same intonation unit in this manner. This contradicts Chafe's account of previously activated ideas (cf. Chafe 1994: 110–111) and assumes a quite high capacity for cognitive load for parties in an interaction. The second explanation is that `075` was also previously activated. The activation would have had to take place through inference on the part of the interlocutors about the goals of the bat, and that these goals were typical. What else would the bat do with an eye that was not to be eaten, except throw it away? This account maintains Chafe's account for previously activated ideas but requires us to assume that a speaker actively predicts that their interlocutors can infer this information.

5.4 Interactional goals

The last factor which actively affects the process of verbalisation is the interactional goals of the speaker. Here, too, we can observe effects of interactional goals on all stages of the verbalisation process which involve construal, with the effects most obvious during the particularising stage. Information-rich intonation units which are primarily explainable via the effects of interactional goals on construal provide evidence for the presence of centres of interest in the verbalisation process and the role of interpersonal dynamics in conversation.

One common strategy found across the texts is the use of presentative constructions. In these instances, what may appear to be an information-rich intonation unit due to the presence of multiple potential discourse-new ideas being encoded can be considered as only communicating a single discourse idea. For example, in (10) from Kera'a, `030` was used to introduce the main character Ano into the discourse for the first time, while in (11), the negative copula corresponding to `013` is used with the first mention of *key ama* 'wax bee'. In both cases, the amount of information associated with the event was minimal. The choice to opt for a presentative strategy can be primarily understood as fulfilling interactive goals, given that a speaker could have just as easily realised the discourse-new referent along with the first action that they carry out.

(10) Kera'a: Dog Story 59

anome lame

ano-me	la-me
Ano-NOM	say-CC
new	new
new	new
	030

'(there was a human) called Ano …'

(11) Duhumbi: Macaque 23

kej ama baŋɲi *, | kej ama . //*

key_ama	bang=nyi	key_ama
beeswax_bee	NEG.EXIST=Q	beeswax_bee
new	new	1
new	new	1-same
	013	

'(wasn't there) the wax bee? The wax bee.'

We also see interactional goals affecting how discourse ideas are construed in cases of contrastive focus. In (12), a continuation of (2) (repeated below for convenience), both the 079 and the referent *chikhus* 'honey' are discourse-new. An explicit contrast is set up between *chikhus* 'honey' and *khum* 'eye' by the use of *boyuba* 'not' in (2). The use of *boyuba* 'not' creates a situation of contrastive focus which "projects" a forthcoming second referent in the next intonation unit (Bodt 2020: 427–428; cf. Auer 2005).[7] As such, an interactant can already anticipate what information may be verbalised in the next intonation unit, namely a referent which is an alternative member of the set of things which can

[7] This analysis of the structure is the most parsimonious with Bodt's original translation of "not the eye, but honey" (Bodt 2018b; cf. Bodt 2020: 426–429). However, a second interpretation is possible, in which *boyuba* 'not' is a finite form and can be interpreted as the end of a syntactic project. In this instance, we may want to suggest that there is information coming from the prosodic signal which sets up the contrastive focus interpretation, i.e., there is an unfinished prosodic project.

be fed. This suggests that both *chikhus* ‘honey’ and *khum* ‘eye’ were both part of the centre of interest of the speaker already at the stage of subchunking but that the speaker chose to verbalise the two ideas in a way which should link these ideas in the audience’s mind. That is, the verbalisation does not just comply with the ONICON, it also leverages this constraint in a way that furthers the interactional goals of the speaker (*pace* Chafe 1987: 38).

(2) Duhumbi: Macaque 94

zok leda kʰum bojuba

zok	le-da	khum	boyu-ba
lie	do-NF	eye	NEG.EQ-NMLZ
new	new	5	new
new	new	2-same	new
	077		078

‘…(the bat) telling that lie, (fed to the macaque) not the eye…’

(12) Duhumbi: Macaque 95

tɕʰikʰus baŋɕibale . //

chikhus	bang-shi-ba=le
honey	feed-give-NMLZ=COP.LE
new	new
new	new+new
	079

‘but fed honey.’

Not all instances of verbalisation for interactional goals must result in separate intonational unit realisations. In (13), *khami* ‘not be’ and *ngadoba* ‘not exist’ are used synonymously to encode the state of ‘nothing more existing’.[8] This pattern can alternatively be analysed as encoding a recognisable pattern of lexical semantic overlap (see Reinöhl & Peck 2026). Here, we can see the same discourse idea being realised with two lexical items as a way for the speaker to emphasise the potential

[8] See Section 5.3 for an account of the discourse-new status of 062 and 063.

destruction that the dog character might cause to the world. In other words, the repetition serves an interactional purpose.

(13) Kera’a: Dog Story 112

kʰam ŋado ba *lajo buda lane*

kha-mi	nga-do-ba	la-o=bu=da	la-ne
lay-NEG	NEG.EXIST-finish-go	say-SUBJ=ADD=DECL	say-CV
new	0	new	new
new	0-syn+2-same	1-same	0-same
061	061	062	063

‘he’ll mess up everything, there’ll be nothing left’

These examples show ways in which speakers creatively choose to verbalise a given subchunk of experience. In some instances, the way a discourse idea is construed evidences that subchunking involves a higher level of organisation than the focus of consciousness, namely the centre of interest (cf. e.g. Pawley & Syder 1983, 2000). In others, it is in the particularising stage that we see the limits on verbalisation leveraged for rhetorical effect.

6. Discussion

While the ONICON predicts the maximal new content which can be realised in an intonation unit, it does not account for the total information that can be expressed. We do not know the limits of verbalisation. Chafe’s account suggests that when multiple discourse-new ideas are present, they are all semi-active for interlocutors (Chafe 1994: 111). This implies that multiple units of accessible information should not co-occur with completely inactive information, which does not necessarily seem to be the case. The amount of information which can be expressed in a single intonation unit must be limited, as too much information expressed in a single focus of consciousness would lead to difficulties in both production and processing. However, there is no indication that this limit lies in the co-occurrence of one inactive idea and multiple semi-active ideas.

Out of the three texts inspected, it is only in Macaque where we find instances of intonation units which may be hitting the upper limit of how much information can be encoded in a single intonation unit. The quotation realised in the second intonation unit in (8) (repeated below) does not co-occur with the quoting event 020 (the bat saying), which is instead realised in the next intonation unit. This is somewhat surprising, as quotation events are regularly realised in the same intonation unit as the quoted information in Duhumbi (cf. Section 5.3). Is the splitting of events across two intonation units a reflection of an upper constraint on information? Or is it simply a decision on the part of the speaker to emphasise the saying event?

(8) Duhumbi: Macaque 31

gomtɕʰen pʰawaŋ | lok ga ***gepuaʔ tɕʰiliŋ dʑakda*** *|| inda inloŋbeʔ*

gomchen_phawang	lok	ga	gepu-aq	chiling	jak-da
bat	friend	1SG	king-GEN	drum	guard-NF
1	4	1	[new]	new	new
new	4-same	pro	new	new	new
					002

in-da	in-long=beq
say-NF	say-PRF=EXIST
new	
3-same	0-same
020	020

‘The bat had said, “friend, I guard the king’s drum.”’

Similarly, we see a sequence of three discourse-new events split across two intonation units in (14). Based on their contentfulness, it is understandable that 028 (bees helping each other) and 030 (bees biting the macaque) are realised in separate intonation units. However, the realisation of 029 (bees coming) in the second intonation unit is somewhat surprising. Based on the original translation, 028 and 029 form a single syntactic unit, and therefore probably a single discourse idea (cf. Bodt 2020: 434–436). This would suggest that a verbalisation of both these events in the same intonation unit would be expected, instead of the separated verbalisation seen in (14). The presence of an intonation break and absence of any other prosodic information between the events suggests that the speaker hit some sort of cognitive limit when uttering this example. The whole subchunk could not be realised in a single intonation unit.

(14) Duhumbi: Macaque 47

kej amabak gaŋpu gaŋpu gaŋpu ***rupta*** | ***londa*** *wojɲi zekda* .

key_ama-bak	gangpu	gangpu	gangpu	rup-ta	lon-da	woy-nyi	zek-da
beeswax.bee-PL	all	all	all	help-NF	come-NF	3SG-DAT	bite-NF
6				new	new	5(2)	new
4-same	pro	pro	pro	new	new	pro	new
				028	029		030

'…each and every single one of the the wax bees, coming to help, stinging him…'

If we broaden our focus past the first four stages of verbalisation, we can also see that certain structures reoccurred across the three texts to construe information-rich intonation units. Multiple new ideas in the same intonation unit were often realised with clause linking strategies or embedding constructions involving finite verbs. This suggests that the stages of structuring and cohering are also inherently affected by the ONICON. Signalling boundaries between clauses explicitly may be a strategy to support easier on-the-fly processing of the information by interlocutors. The preference for explicitly signalling clausal boundaries when multiple units of information are expressed in the same prosodic unit also suggests that there is some validity to the argument that clauses typically express the verbalisation of events (cf. Croft 2007; Thompson & Couper-Kuhlen 2005). Experimental studies which specifically target information-rich intonation units may be able to help us further ascertain the role exact cognitive mechanisms underlying the ONICON play.

Despite the theoretical implications of this study, the arguments so far crucially rely on the assumption that the ONICON acts as a hard constraint on cognition. However, it is not a given that the ONICON is an absolute universal. The ONICON may feasibly represent a cultural preference for how much information we like to package in single intonation units and exist only as a socio-cultural practice in the interactive realm. Information from first language acquisition processes may be able to illuminate what is more likely to be true, given that the latter explanation relies on an active process of learning a preference rather than an underlying constraint on memory. Certainly, processes affecting construal are to some extent socioculturally-determined (cf. e.g. Pawley 1987) and some are also developmentally-sensitive (e.g. Grigoroglou & Papafragou 2019). A weakening of the constraint does, however, result in a loss of the ONICON's predictive power and a reduced need to account for cases like information-rich intonation units which might otherwise prove beneficial to refining the constraint.

Furthermore, the previous discussion does not take a number of properties of interaction into consideration. For example, gestural cues have been demonstrated to aid in negotiation and comprehension of meaning in discourse (Dargue, Sweller & Jones 2019). What would we find when we include multimodal cues in an account of verbalisation? Would visual cues be present alongside more information-rich units or would this result in an information overload? Additionally, the ONICON and verbalisation as operationalised here only account for contentful, or "substantive", intonation units, despite interactional moves such as repairs or reassurance also frequently occurring in conversations. Intonation units which express these sorts of "regulatory" moves are not considered by Chafe as encoding ideas, and as such, he argues that the ONICON does not apply in these cases (e.g. Chafe 1994: 64–65, 108). However, we can find instances of information-rich intonation units which are used for regulatory purposes, such as (15). In this example, *alrə* 'good' was used by the speaker to express basic reassurance about the event encoded by `003` (talking). The verbalisation process which resulted in these two discourse-new pieces of information being realised in the same intonation unit was clearly affected by interactional goals of the speaker in this instance. Even if the role of the uttered intonation unit is regulatory, it is highly likely that the same processes and constraints apply to how the unit is verbalised. As such, more attention needs to be paid to naturalistic data from interactions in a range of genres and languages to be able to see what principles truly underlie how we choose to structure discourse.

(15) Galo: Lost Writing 4

əna no ***mentᵊ ke alrə***

əə=na	nó	mèn-tó	kèe	alə́-rə́
ANA=DECL	2SG	speak-IPFV	HORT.POL	good-IRR
	SAP	new		new
	pro	new		new
		003		004

‘That’s right, go ahead talking, it’ll be ok.’

Other approaches to capturing “ideahood” may also capture information flow principles better than the ONICON does. For example, Lambrecht (1994)’s focus-presupposition account may be more insightful for capturing the dynamics of information flow in conversation.[9] In this approach, a clausal focus appears to be roughly equal to Chafe’s “one new idea”. Notably, Lambrecht’s sentence focus would be able to account for information-rich intonation units in which all information is new, where a purely Chafian account struggles (cf. examples in Ozerov 2026; Mithun 2026). A proper test of this account would however first require Lambrecht’s approach to be applied on an intonation unit level, as argued by Reinöhl & Peck (2026).

The ONICON as a universal on human cognitive capabilities is only now beginning to undergo detailed empirical scrutiny. This state of affairs also entails that we do not yet know much about the main cognitive process with which it interacts, namely verbalisation. What is evident, however, is that information-rich intonation units are natural consequences of how language is used, not exceptions. By understanding how and why these units may occur in naturalistic language, we grow ever slightly closer to understanding processes of human cognition and the effects that these processes have on the patterns of language that we see every day.

References

Auer, Peter. 2005. Projection in interaction and projection in grammar. *Text - Interdisciplinary Journal for the Study of Discourse* 25(1). 7–36. https://doi.org/10/csnpkk.

[9] Many thanks to the anonymous reviewer who brought the cognitive richness of Lambrecht’s work to my attention.

Baruah, Urmimala & Ruma Bhattacharyya. 2019. Diverse ethnic food practices of the Galo tribe in Arunachal Pradesh. *Food Science Research Journal* 10(2). 197–202. https://doi.org/10.15740/HAS/FSRJ/10.2/197-202.

Bodt, Timotheus Adrianus. 2018a. Duhumbi Stories - Sound files. Zenodo. https://doi.org/10.5281/ZENODO.1400495.

Bodt, Timotheus Adrianus. 2018b. Duhumbi Stories - Transcribed, parsed, glossed, translated text files. Zenodo. https://doi.org/10.5281/ZENODO.1400505.

Bodt, Timotheus Adrianus. 2020. *Grammar of Duhumbi (Chugpa)*. Leiden: Brill.

Chafe, Wallace. 1974. Language and consciousness. *Language* 50(1). 111–133. https://doi.org/10.2307/412014.

Chafe, Wallace. 1976. Givenness, contrastiveness, definiteness, subjects, topics, and point of view. In Charles N. Li (ed.), *Subject and topic*, 25–56. New York: Academic Press.

Chafe, Wallace. 1977. Creativity in verbalization and its implications for the nature of stored knowledge. In Roy O. Freedle (ed.), *Discourse production and comprehension*, 41–55. Norwood: Ablex.

Chafe, Wallace. 1979. The flow of thought and the flow of language. In Talmy Givón (ed.), *Discourse and syntax*, 159–181. New York: Academic Press.

Chafe, Wallace. 1987. Cognitive constraints on information flow. In Russell S. Tomlin (ed.), *Coherence and grounding in discourse: Outcome of a symposium, Eugene, Oregon, June 1984*, 21–52. Amsterdam: John Benjamins.

Chafe, Wallace. 1994. *Discourse, consciousness, and time: The flow and displacement of conscious experience in speaking and writing*. Chicago: University of Chicago Press.

Chafe, Wallace. 1998. Things we can learn from repeated tellings of the same experience. *Narrative Inquiry* 8(2). 269–285. https://doi.org/10.1075/ni.8.2.03cha.

Chafe, Wallace. 2018. *Thought-based Linguistics*. Cambridge: Cambridge University Press.

Christiansen, Morten H. & Nick Chater. 2016. Language as Shaped by the Brain. In *Creating Language: Integrating Evolution, Acquisition, and Processing*, 19–65. Cambridge: MIT Press. http://www.jstor.org/stable/j.ctt1b7x5tw.6.

Clark, Herbert H. 1996. *Using language*. Cambridge: Cambridge University Press.

Croft, William. 1995. Intonation units and grammatical structure. *Linguistics* 33(5). 839–882. https://doi.org/10/bzr326.

Croft, William. 2007. The origins of grammar in the verbalization of experience. *Cognitive Linguistics* 18(3). https://doi.org/10.1515/COG.2007.021.

Damron, Rebecca. 2004. Prosodic schemas: Evidence from Urdu and Pakistani English. In Carol Lynn Moder & Aida Martinovic-Zic (eds.), *Discourse across languages and cultures*, 53–73. Amsterdam: John Benjamins.

Dargue, Nicole, Naomi Sweller & Michael P. Jones. 2019. When our hands help us understand: A meta-analysis into the effects of gesture on comprehension. *Psychological Bulletin* 145(8). 765–784. https://doi.org/10.1037/bul0000202.

Delin, Judy. 1996. Review of Chafe (1994): Discourse, Consciousness, and Time: The Flow and Displacement of Conscious Experience in Speaking and Writing. *Functions of Language* 3(1). 129–133. https://doi.org/10.1075/fol.3.1.08del.

Dunbar, R. I. M. 2004. Gossip in evolutionary perspective. *Review of General Psychology*. SAGE Publications 8(2). 100–110. https://doi.org/10.1037/1089-2680.8.2.100.

Ericsson, K. Anders & Herbert A. Simon. 1980. Verbal reports as data. *Psychological Review*. US: American Psychological Association 87(3). 215–251. https://doi.org/10.1037/0033-295X.87.3. 215.

Ericsson, K. Anders & Herbert A. Simon. 1998. How to study thinking in everyday life: contrasting think-aloud protocols with descriptions and explanations of thinking. *Mind, Culture, and Activity* 5(3). 178–186. https://doi.org/10.1207/s15327884mca0503_3.

Evans, Nicholas, Henrik Bergqvist & Lila San Roque. 2018. The grammar of engagement I: Framework and initial exemplification. *Language and Cognition* 10(1). 110–140. https://doi.org/10/gdmxnq.

Evans, Nicholas & Stephen C. Levinson. 2009. The myth of language universals: Language diversity and its importance for cognitive science. *Behavioral and Brain Sciences* 32(5). 429–448. https://doi.org/10/czvgpd.

Givón, T. (ed.). 1983. *Topic continuity in discourse: A quantitative cross-language study*. Amsterdam: John Benjamins.

Givón, T. 1991. Some substantive issues concerning verb serialization: Grammatical vs. cognitive packaging. In Claire Lefebvre (ed.), *Serial Verbs: Grammatical, Comparative and Cognitive Approaches*, 137–184. Amsterdam: John Benjamins.

Graesser, Arthur C., Murray Singer & Tom Trabasso. 1994. Constructing inferences during narrative text comprehension. *Psychological Review* 101(3). 371–395. https://doi.org/10.1037/0033-295X.101.3.371.

Grice, H. Paul. 1975. Logic and conversation. In Peter Cole & Jerry L. Morgan (eds.), *Syntax and semantics 3: Speech arts*, 41–58. New York: Academic Press.

Grigoroglou, Myrto & Anna Papafragou. 2019. Interactive contexts increase informativeness in children's referential communication. *Developmental Psychology*. American Psychological Association (APA) 55(5). 951–966. https://doi.org/10.1037/dev0000693.

Hopper, Paul J. 1995. The Category "Event" in Natural Discourse and Logic. In Werner Abraham, T. Givón & Sandra A. Thompson (eds.), *Discourse, Grammar and Typology: Papers in honor of John W.M. Verhaar*, 139. Amsterdam: John Benjamins. https://doi.org/10.1075/slcs.27.12hop.

Johnson-Laird, Philip Nicholas. 1983. *Mental models: Towards a cognitive science of language, inference, and consciousness*. Cambridge: Harvard University Press.

Lambrecht, Knud. 1994. *Topic, focus, and the mental representations of discourse referents*. Cambridge: Cambridge University Press.

Langacker, Ronald. 1991. *Descriptive application*. Vol. 2. Stanford: Stanford University Press.

Levelt, Willem J. M. 1989. *Speaking: From Intention to Articulation*. Cambridge: The MIT Press. https://doi.org/10.7551/mitpress/6393.001.0001.

Linders, Guido M. & Stefan Schnell. 2026. A cross-linguistic computational study on One New Idea per clause. *Functions of Language*.

Magliano, Joseph P. & Gabriel A. Radvansky. 2001. Goal coordination in narrative comprehension. *Psychonomic Bulletin & Review* 8(2). 372–376. https://doi.org/10.3758/BF03196175.

Malt, Barbara C., Steven A. Sloman, Silvia Gennari, Meiyi Shi & Yuan Wang. 1999. Knowing versus naming: similarity and the linguistic categorization of artifacts. *Journal of Memory and Language* 40(2). 230–262. https://doi.org/10.1006/jmla.1998.2593.

Mithun, Marianne. 1987. Is basic word order universal? In Russell S. Tomlin (ed.), *Coherence and grounding in discourse: Outcome of a symposium, Eugene, Oregon, June 1984*, 281–328. Amsterdam: Benjamins.

Mithun, Marianne. 2026. The role of information packaging in shaping linguistic structure. *Functions of Language*.

Ozerov, Pavel. 2026. Beyond a single constraint: The "one-new-idea" tendency as an emergent phenomenon. *Functions of Language*.

Papafragou, Anna & Myrto Grigoroglou. 2019. The role of conceptualization during language production: Evidence from event encoding. *Language, Cognition and Neuroscience* 34(9). 1117–1128. https://doi.org/10/gmmdfq.

Pawley, Andrew. 1987. Encoding events in Kalam and English: Different logics for reporting experience. In Russell S. Tomlin (ed.), *Coherence and grounding in discourse: Outcome of a symposium, Eugene, Oregon, June 1984*, 329–360. Amsterdam: John Benjamins.

Pawley, Andrew. 2011. Event representation in serial verb constructions. In Jürgen Bohnemeyer & Eric Pederson (eds.), *Event Representation in Language and Cognition*, 13–42. Cambridge: Cambridge University Press.

Pawley, Andrew & Frances Hodgetts Syder. 1983. Two puzzles for linguistic theory: Nativelike selection and nativelike fluency. In Jack C. Richards & Richard W Schmidt (eds.), *Language and communication*, 191–225. London: Longman.

Pawley, Andrew & Frances Hodgetts Syder. 2000. The one-clause-at-a-time hypothesis. In Heidi Riggenbach (ed.), *Perspectives on Fluency*, 163–199. Ann Arbor: The University of Michigan Press.

Peck, Naomi. 2025a. *(Re)contextualising serialisation: Multiverbal constructions in the East Himalaya*. Freiburg: University of Freiburg PhD thesis. https://doi.org/10.6094/UNIFR/ 274498.

Peck, Naomi. 2025b. Kera'a Corpus. OSF. https://doi.org/10.17605/OSF.IO/UX6PV.

Pickering, Martin J. & Simon Garrod. 2013. An integrated theory of language production and comprehension. *Behavioral and Brain Sciences* 36(4). 329–347. https://doi.org/10.1017/S0140525X12001495.

Post, Mark W. 2006. Loss of writing. PARADISEC. https://doi.org/10.26278/5D275D3752D15.

Radvansky, Gabriel A. & Jeffrey M. Zacks. 2014. *Event cognition*. Oxford: Oxford University Press.

Reinöhl, Uta, Kirsten Culhane, Simon Fries, Naomi Peck & Maria Vollmer. 2023. "One new idea" constraint holds cross-linguistically even in "flat" expressions. Oral Presentation presented at the 16th International Cognitive Linguistics Conference, Heinrich-Heine-Universität Düsseldorf.

Reinöhl, Uta & T. Mark Ellison. 2026. The one-new-idea constraint, functor-argument metaphors, and holistic language processing. *Functions of Language*.

Reinöhl, Uta & Naomi Peck. 2026. The 'idea' in the One-New-Idea Constraint. *Functions of Language*.

Slobin, Dan I. 1996. From "thought and language" to "thinking for speaking". In J. J. Gumperz & Stephen C. Levinson (eds.), *Rethinking linguistic relativity*, 70–96. Cambridge: Cambridge University Press.

Stathi, Katerina. 2023. *Granularity in the Verbalization of Events and Objects: A cross-linguistic study*. Amsterdam: John Benjamins. https://doi.org/10.1075/slcs.233.

Swallow, Khena M., Jeffrey M. Zacks & Richard A. Abrams. 2009. Event boundaries in perception affect memory encoding and updating. *Journal of Experimental Psychology: General* 138(2). 236–257. https://doi.org/10.1037/a0015631.

Szala, Anna, Sławomir Wacewicz, Marek Placiński, Aleksandra Ewa Poniewierska, Arkadiusz Schmeichel, Michał Stefańczyk, Przemyslaw Zywiczynski & R. I. M. Dunbar. 2024. How much language use is actually on social topics: Human conversational behavior revisited. https://doi.org/10.31234/osf.io/d42gr.

Tenbrink, Thora. 2015. Cognitive Discourse Analysis: Accessing cognitive representations and processes through language data. *Language and Cognition* 7(1). 98–137. https://doi.org/10.1017/langcog.2014.19.

Thompson, Sandra A. & Elizabeth Couper-Kuhlen. 2005. The clause as a locus of grammar and interaction. *Discourse Studies* 7(4–5). 481–505. https://doi.org/10.1177/1461445605054403.

Trueswell, John C. & Anna Papafragou. 2010. Perceiving and remembering events cross-linguistically: Evidence from dual-task paradigms. *Journal of Memory and Language*. Elsevier BV 63(1). 64–82. https://doi.org/10.1016/j.jml.2010.02.006.

Ünal, Ercenur, Yue Ji & Anna Papafragou. 2021. From event representation to linguistic meaning. *Topics in Cognitive Science* 13(1). 224–242. https://doi.org/10/ghtv77.

Vogels, J., D. M. Howcroft, E. Tourtouri & V. Demberg. 2020. How speakers adapt object descriptions to listeners under load. *Language, Cognition and Neuroscience* 35(1). 78–92.

Wittgenstein, Ludwig. 1953. *Philosophical investigations*. Oxford: Blackwell.

Yule, George. 1981. New, current and displaced entity reference. *Lingua*. Elsevier BV 55(1). 41–52. https://doi.org/10.1016/0024-3841(81)90054-1.

Zacks, Jeffrey M. & Khena M. Swallow. 2007. Event segmentation. *Current Directions in Psychological Science* 16(2). 80–84. https://doi.org/10.1111/j.1467-8721.2007.00480.x.

Naomi Peck

npeck@uni-bremen.de

University of Bremen

Fachbereich 10: Sprach- und Literaturwissenschaften

Universitäts-Boulevard 13, Gebäude GW 2

28359 Bremen

Germany